# CLOUD COMPUTING

## Introduction to Cloud Computing
## Terminologies (Cont.)
## Grid & Cloud Computing

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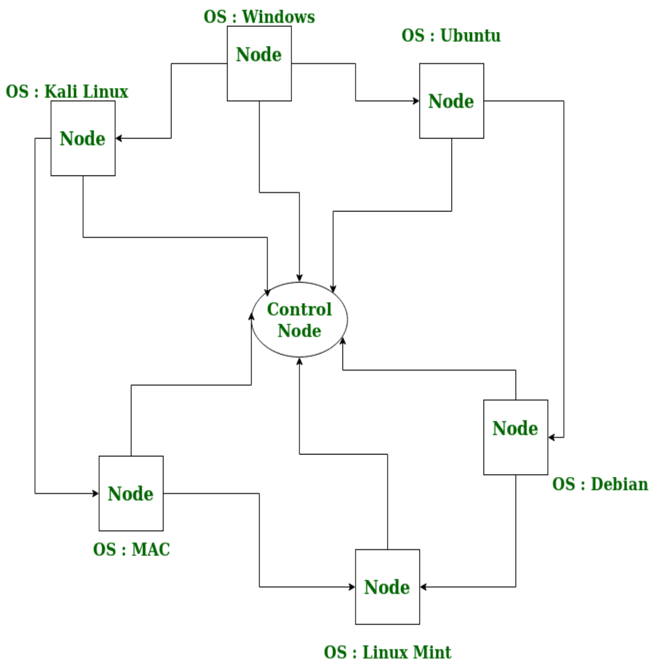
# Evolution of Computing : Grid Computing

Grid computing is defined as *"coordinated resource sharing and problem solving in dynamic, multi-institutional virtual organizations."*

- Grid computing is the use of widely distributed computer resources to reach a common goal.
- A computing grid can be thought of as a distributed system with non-interactive workloads that involve many files.
- Grid computing is distinguished from conventional high-performance computing systems such as cluster computing in that grid computers have each node set to perform a different task/application.
- Grid computers also tend to be more heterogeneous and geographically dispersed (thus not physically coupled) than cluster computers
- Establishing **trust and security models** between infrastructure resources pooled from two different administrative domains become even more important.
- **Resource sharing agreements** needed to be formed among a set of participating parties where sharing meant **DIRECT ACCESS** to the resources.
- Sharing was **highly controlled** with resource providers and consumers grouped into virtual organizations primarily based on sharing conditions.




Topology in Grid Computing

(https://en.wikipedia.org/wiki/Grid_computing)

https://www.geeksforgeeks.org/grid-computing/

# CLOUD COMPUTING

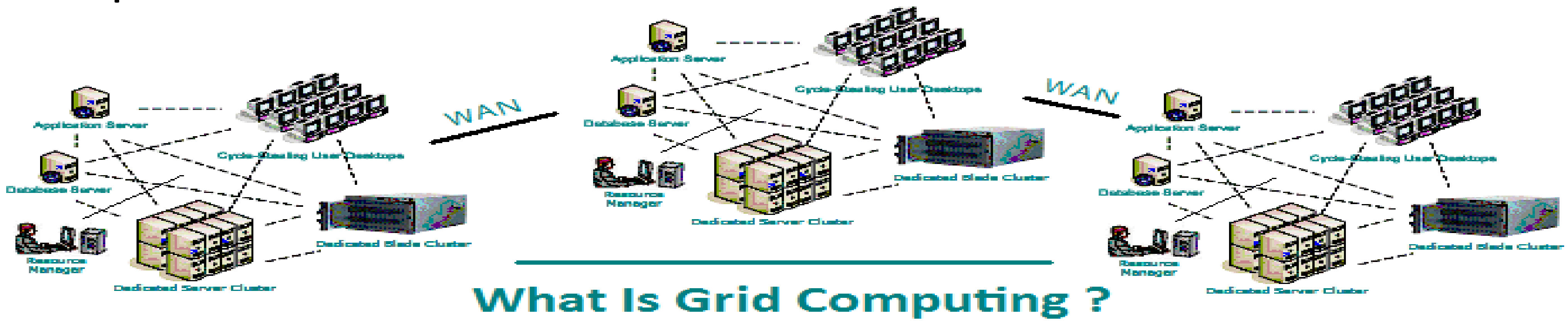
## Grid Computing

Cloud computing is frequently compared to grid computing but here are the features of Grid which constrasts.

- The vision of grid computing is to enable **computing to be delivered as a utility**
- grid computing was meant to be used by individual users who gain access to computing devices **without knowing** where the **resource is located** or **what hardware it is running**, and so on.
- Initial technological focus of grid computing was **limited to enabling shared use of resources with common protocols for access**

**Features of a Grid**

1. Co-ordinates resources that are not subject to centralized control
2. Using standard, open, general purpose protocols and interfaces
3. To deliver non-trivial quality of service
4. Uses a common standard for authentication, authorization, resource discovery and resource access

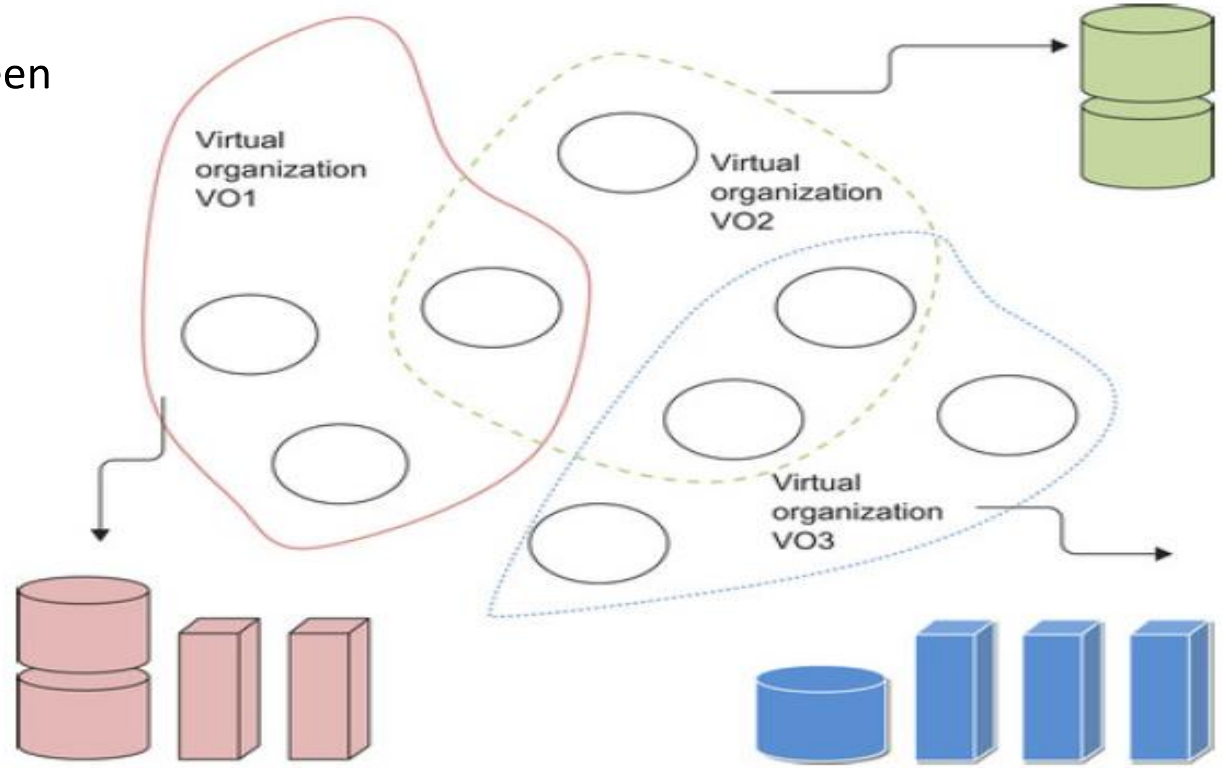





https://www.esds.co.in/kb/what-is-grid-computing/

# Grid Computing

- Uses the **Concept of Virtual Organization (VO)** which is basically a dynamic collection of individuals or institutions from multiple administrative domains. It is used to enable flexible, coordinated, secure resource sharing among participating entities (T2)
- A VO forms a **basic unit for enabling access to shared resources** with specific resource-sharing policies applicable for users from a particular VO
- Grid Technology enables sharing of resource between parties who may not have any trust earlier




**Benefits**

- Exploit Underutilized resources
- Resource load Balancing
- Virtualize resources across an enterprise
- Data Grids, Compute Grids
- Enable collaboration for virtual organizations

## Grid Computing : Computational and Data Grid

"A computational grid is a **hardware** and **software** infrastructure that provides dependable, consistent, pervasive, and inexpensive access to **high-end computational capabilities**."

Example : Science Grid (US Department of Energy)

A **data grid** is the storage component of a grid computing system that deals with the **controlled sharing and management of large amounts of distributed data**.

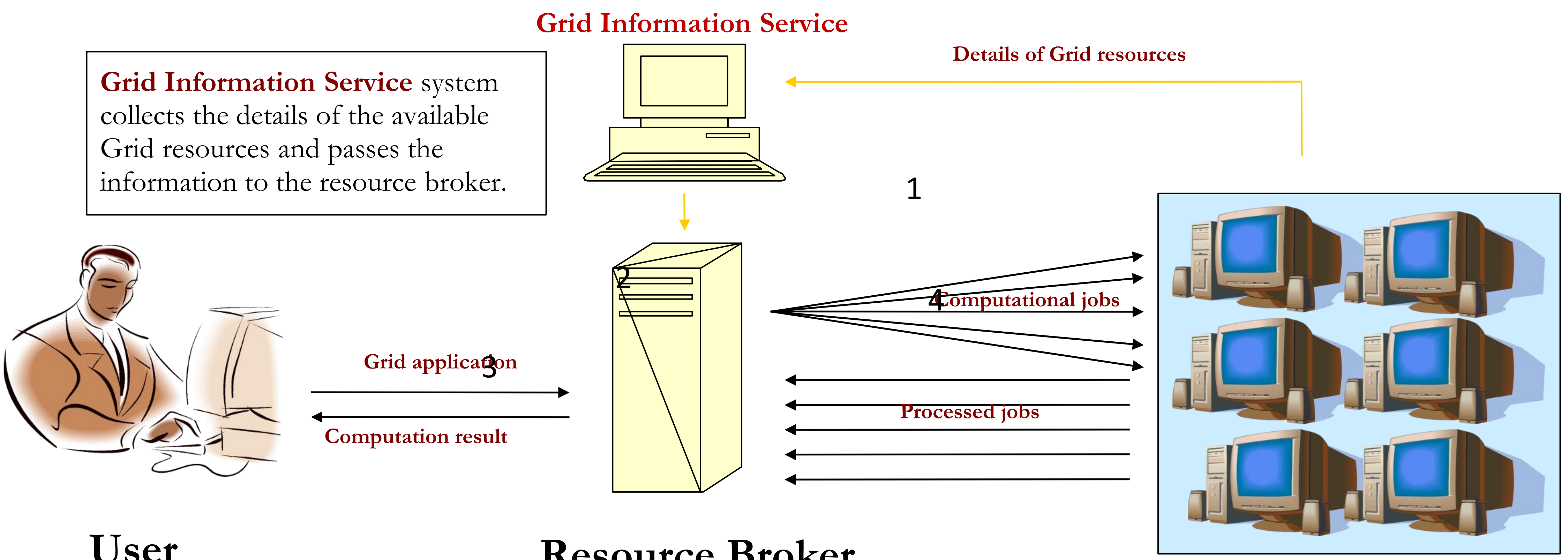
Scientific and engineering applications require access to large amounts of data, and often this data may be widely distributed. A data grid provides seamless access to the local or remote data required to complete compute intensive calculations.

Examples of Data Grid's :

Biomedical informatics Research Network (BIRN),

Southern California earthquake Center (SCEC).

# Grid Computing : A typical view of the Grid environment




## User

A **User** sends computation or data intensive application to Global Grids in order to speed up the execution of the application.

## Resource Broker

A **Resource Broker** distribute the jobs in an application to the Grid resources based on user's QoS requirements and details of available Grid resources for further executions.

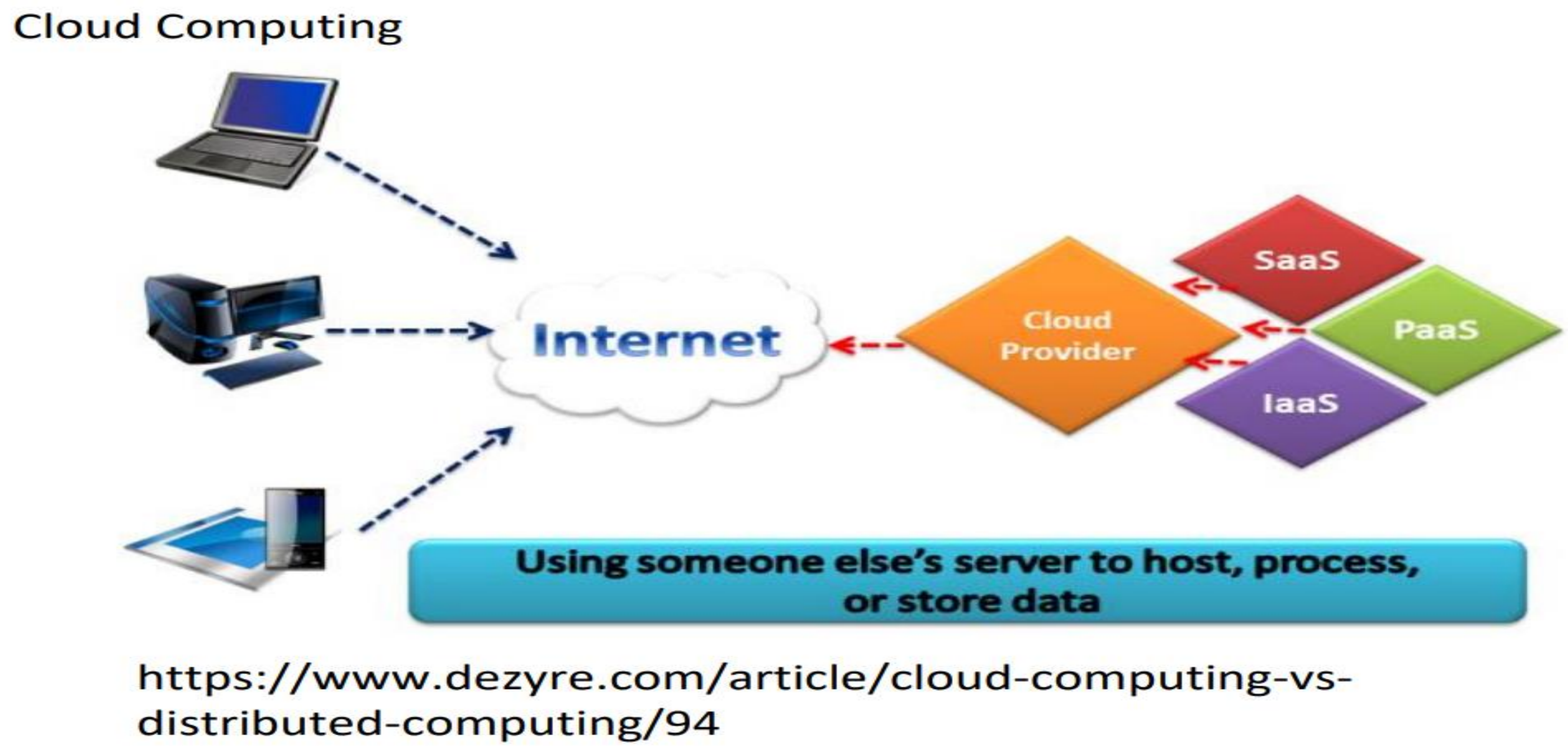
## Grid Resources

**Grid Resources** (Cluster, PC, Supercomputer, database, instruments, etc.) in the Global Grid execute the user jobs.

## Evolution of Computing : Cloud Computing

**Cloud Computing**

An Internet cloud of resources can be either a centralized or a distributed computing system. The cloud applies parallel or distributed computing, or both. Clouds can be built with physical or virtualized resources over large data centers that are centralized or distributed.



### Applications of Cloud Computing (T1 1.1.2.2)

| Domain | Specific Applications |
|---|---|
| Science and engineering | Scientific simulations, genomic analysis, etc. |
| | Earthquake prediction, global warming, weather forecasting, etc. |
| Business, education, services industry, and health care | Telecommunication, content delivery, e-commerce, etc. |
| | Banking, stock exchanges, transaction processing, etc. |
| | Air traffic control, electric power grids, distance education, etc. |
| | Health care, hospital automation, telemedicine, etc. |
| Internet and web services, and government applications | Internet search, data centers, decision-making systems, etc. |
| | Traffic monitoring, worm containment, cyber security, etc. |
| | Digital government, online tax return processing, social networking, etc. |
| Mission-critical applications | Military command and control, intelligent systems, crisis management, etc. |

# Cloud Computing Design Objectives

## Design objectives of cloud computing

- **Efficiency** measures the utilization rate of resources in an execution model by exploiting massive parallelism in HPC. For HTC, efficiency is more closely related to job throughput and power efficiency. All of these at lowest cost.
- **Dependability** measures the reliability and self-management from the chip to the system and application levels. The purpose is to provide high-throughput service with Quality of Service (QoS) assurance, even under failure conditions.
- **Adaptation** in the programming model measures the ability to support billions of job requests over massive data sets and virtualized cloud resources under various workload and service models.
- **Flexibility** in application deployment measures the ability of distributed systems to run well in both HPC (science and engineering) and HTC (business) applications.

# Cloud Computing - Evolution

**Evolution of the Web ( A key enabler in cloud computing )**

**Web 1.0**

Four design essentials of a Web 1.0 site include:

1.Static pages.

2.Content is served from the server’s file-system.

3.Pages built using Server Side Includes or Common Gateway Interface (CGI).

4.Frames and Tables used to position and align the elements on a page.

**Web 2.0 –**

Web 2.0 refers to world wide website which highlight user-generated content, usability &interoperability for end users. Web 2.0 is also called participative social web. Ajax & JavaScript frameworks extensively used in Web 2.0

**Major features of Web 2.0 –**

1. Permits users to retrieve and classify the information collectively.
2. Dynamic content that is responsive to user input.
3. Information flows between site owner and site users by means of evaluation & online commenting.
4. Developed APIs to allow self-usage, such as by a software application.

## Web 3.0

**1.Semantic Web**

Generate, share and connect content through search and analysis based on **the ability to understand the meaning of words**, rather than on keywords or numbers.

**2.Artificial Intelligence**

Combining this capability with natural language processing, in Web 3.0,

**3.3D Graphics**

**The three dimensional design is being used extensively in websites and services** in Web 3.0. Museum guides, computer games, ecommerce, geospatial contexts, etc. are all examples that use 3D graphics.

**4.Ubiquity**

**Content is accessible by multiple applications**, every device is connected to the web, the services can be used everywhere.

# Contrasting Cloud Computing and Grid Computing

| Cloud Computing | Grid Computing |
|---|---|
| Cloud Computing follows client-server computing architecture. | Grid computing follows a distributed computing architecture. |
| Scalability is high. | Scalability is normal. |
| Cloud Computing is more flexible than grid computing. | Grid Computing is less flexible than cloud computing. |
| Cloud operates as a centralized management system. | Grid operates as a decentralized management system. |
| In cloud computing, cloud servers are owned by infrastructure providers. | In Grid computing, grids are owned and managed by the organization. |
| Cloud computing uses services like Iaas, PaaS, and SaaS. | Grid computing uses systems like distributed computing, distributed information, and distributed pervasive. |
| Cloud Computing is Service-oriented. | Grid Computing is Application-oriented. |
| It is accessible through standard web protocols. | It is accessible through grid middleware. |

https://www.javatpoint.com/cloud-computing-vs-grid-computing

## Compute as an Utility ( T1 1.1.2.3)

Utility computing focuses on a business model in which customers receive computing resources from a paid service provider. All grid/cloud platforms are regarded as utility service providers.

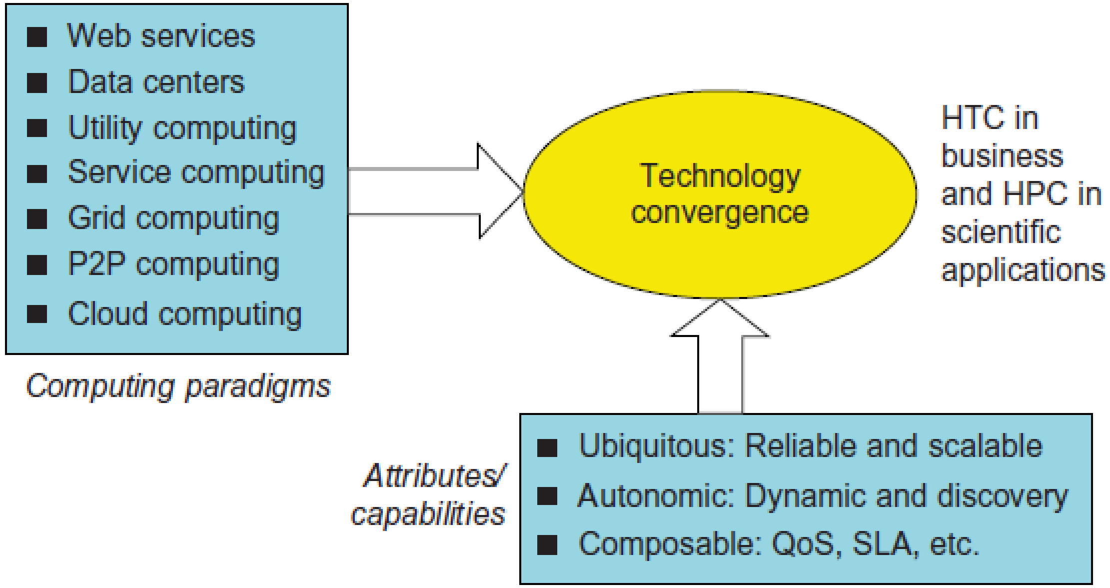



**FIGURE 1.2**

The vision of computer utilities in modern distributed computing systems.

*(Modified from presentation slide by Raj Buyya, 2010)*

# THANK YOU

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